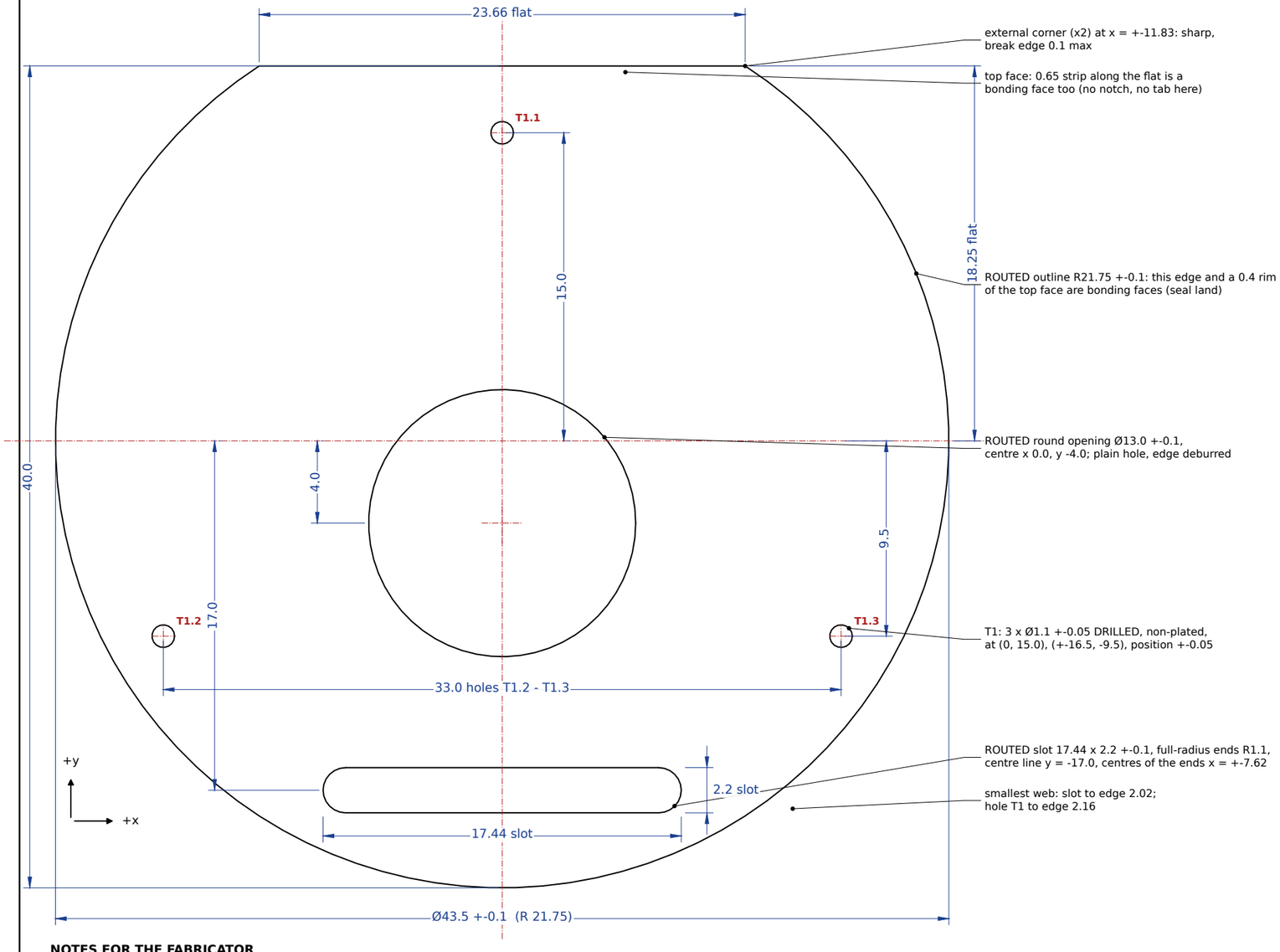
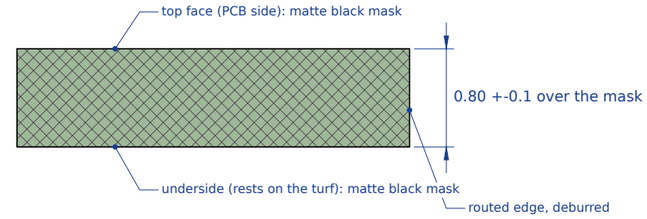
VIEW ON THE TOP FACE (PCB side), z = 0.80 4:1 datum: centre of the R 21.75 arc, +y toward the flat



EDGE VIEW (section y = 0 at the +x edge) 20:1

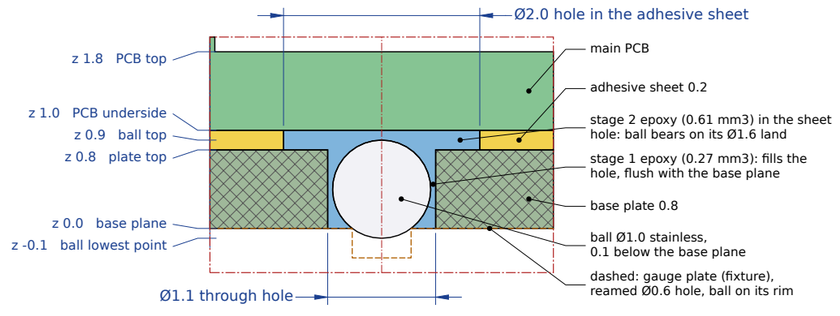


Tool / feature	Qty	Size	Tolerance	Position (x, y)	Plating
T1 drill	3	Ø1.1	+-0.05 dia, +-0.05 pos.	(0, 15.0), (+-16.5, -9.5)	none (NPTH)
Rout: round opening	1	Ø13.0	+-0.1	(0.0, -4.0)	-
Rout: slot	1	17.44 x 2.2, ends R1.1	+-0.1	centre (0, -17.0)	-
Rout: outline	1	R21.75, flat at y = 18.25	+-0.1	arc centre (0, 0)	-
Mask relief at T1	3 + 3	Ø1.25 opening, both faces	+-0.05	on the T1 holes	-
Thickness over mask	-	0.8	+-0.1	-	no copper

Web / edge distance (measured on the model)	mm
hole T1.1 to the outline	2.7
hole T1.2 to the outline	2.16
hole T1.3 to the outline	2.16
round opening to the outline	11.25
slot to the outline	2.02
slot to the round opening	5.4

Smallest internal radius: R1.1 (slot ends). No internal corner below R0.5; no drilled relief needed. A router bit of Ø2.0 or smaller makes every feature. Smallest web 2.02 mm.

FOR INFORMATION (not a fabrication detail): hole T1.1 in use 16:1



NOTES FOR THE FABRICATOR

1. MATERIAL: FR-4 glass-epoxy laminate, NO COPPER on any layer (unclad or fully etched): a mechanical plate, not a circuit. FINISHED THICKNESS OVER THE SOLDER MASK 0.8 +-0.1 mm. Choose the laminate for that finished thickness and state the build-up in the quote (a 0.8 mm copper-clad laminate ends near 0.75 without its copper). We use plates from 0.76 to 0.90 mm.
2. SOLDER MASK: matte black, both faces, full coverage EXCEPT a round relief Ø1.25 at each hole T1 on both faces: no mask on the hole wall or rim (or drill after mask). NO legend, date code or marking.
3. OUTLINE AND CUT-OUTS: routed, +-0.1. No V-score. Supply individually routed pieces. If panelised: break-off tabs ONLY on the flat (y = 18.25), mouse bites recessed into the outline by at most 0.25 mm, no nub proud of the outline anywhere. NO tab on the round part: it drops into a bore with 0.25 mm radial clearance and carries a 0.4 mm seal land.
4. HOLES T1: 3 x Ø1.1 +-0.05, drilled, non-plated, free of mask; position +-0.05 to each other, +-0.1 to the outline. They take Ø1.0 steel balls (detail, for information).
5. CORNERS: slot ends full radius R1.1; no internal corner below R0.5. The two external corners at the ends of the flat are sharp (break edge 0.1 max).
6. FLATNESS: bow and twist 0.5 % maximum (0.2 mm over the part). Edges free of burrs and loose fibres. Both faces are bonding faces: no release agent, mask fully cured, parts clean.
7. DATA (mm, same origin): drawings/fab/GL1-011_base_plate_FR4-Profile.gbr (Gerber: outline + routed cut-outs), F_Mask.gbr / B_Mask.gbr (mask layers), NPTH.drl (Excellon, NPTH); drawings/dxf/GL1-011_base_plate_FR4_fab.dxf (layers OUTLINE_ROUT, ROUT_OPENING_D13p0, ROUT_SLOT_17p44x2p2, DRILL_NPTH_D1p10); export/step/base_plate.step.
8. Datum: origin = centre of the R21.75 arc; +x to the right and +y toward the flat in this view (seen on the top face). The part is symmetric about the y axis.
9. To report with the first articles: thickness over the mask, hole diameters, outline. PROVISIONAL - not released. Nominal CAD dimensions (design/master_params.yaml rev D); checked by check 15 of the build.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE

GL1-011 Base plate, FR-4 - FABRICATION DRAWING

PROJECT / PART		
GreenLine GL-1 electronic ball marker - base plate (primary build)		
MATERIAL / FINISH	PROCESS	
FR-4 0.80 mm, matte black solder mask both sides, NO COPPER	PCB fabricator: routed outline + cut-outs, drilled NPTH	
GENERAL TOLERANCE	SCALE (A3) / UNITS	
routed +-0.1, drilled +-0.05, thickness +-0.1	4:1 / 20:1 mm	
STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1 / D-EVT1	1 of 1 2026-11-11